

Yuhui Hong

Department of Genome Science
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EDUCATION

Indiana University Bloomington

Ph.D. in Computer Science

Bloomington, IN, US

Sep. 2020 –Jul. 2025

Xidian University

B.S. in Computer Science and Technology

Xi'an, Shaanxi, China

Sep. 2015–Jul. 2019

RESEARCH INTERESTS

My research develops machine learning methods for understanding molecular systems from high-throughput experimental data. I focus on mass spectrometry-based proteomics and metabolomics, and related analytical platforms such as chromatography. My work integrates predictive modeling with interpretable approaches to uncover molecular mechanisms and enable data-driven biological and chemical discovery.

RESEARCH EXPERIENCE

University of Washington

Postdoc Scholar

Advisor: Professor William Stafford Noble

Area: Computational biology, Deep Learning, Proteomics mass spectrometry

Seattle, WA, US

Aug. 2025 –Present

Indiana University Bloomington

Research Assistant

Advisor: Professor Haixu Tang

Area: Computational biology, Deep Learning, Small molecular mass spectrometry

Bloomington, IN, US

Aug. 2021 –Jul. 2025

The First Affiliated Hospital of Nanchang University

Research Intern

Advisor: Professor Sujun Li

Area: MHC (Major Histocompatibility Complex) binding prediction

Nanchang, Jiangxi, China

May 2021 –Jul. 2021

Xi'an Jiaotong University

Research Assistant

Advisor: Associate Professor Yaochen Li

Area: Computer vision, Object tracking and segmentation in traffic scene

Xi'an, Shaanxi, China

Sep. 2019 –Jul. 2020

PUBLICATIONS

The deep learning models, 3DMolMS for retention time and collision-cross section [7], 3DMolCSP for enantioselectivity [5], and FIDDLE for chemical formula identification [3] have been evaluated on internal data and positively considered for application by biotech, pharmaceutical, and agricultural leaders, including **AbbVie**, **Agilent**, **Amgen**, **Merck**, and **Corteva**.

BOOKS & PATENTS

- 1 Qingyang Xiao, Kaiyuan Liu, **Yuhui Hong** & Haixu Tang (2024). "Neural Networks for Chemists." *American Chemical Society*, DOI:10.1021/acsinfocus.7e8012. [Primer]
- 2 Haixu Tang, **Yuhui Hong**, & Sujun Li. "Method of predicting ms/ms spectra and properties of chemical compounds." US Patent No. WO2023239720A1, June 6, 2023.

PEER-REVIEWED ARTICLES (FIRST AUTHOR)

*equal contribution as co-first authors

- 3 **Yuhui Hong**, Sujun Li, Yuzhen Ye, & Haixu Tang (2025). “FIDDLE: a deep learning method for chemical formulas prediction from tandem mass spectra.” *Nature Communications*, 16(1), 11102. [\[Paper\]](#) [\[Code\]](#) [\[PyPI package\]](#)
- 4 **Yuhui Hong**, Yuzhen Ye & Haixu Tang (2025). “Machine Learning in Small-Molecule Mass Spectrometry.” *Annual Review of Analytical Chemistry*, 18. [\[Paper\]](#)
- 5 **Yuhui Hong**, Christopher J Welch, Patrick Piras, & Haixu Tang (2024). “Enhanced Structure-Based Prediction of Chiral Stationary Phases for Chromatographic Enantioseparation from 3D Molecular Conformations.” *Analytical Chemistry*, 96(6), 2351-2359. [\[Paper\]](#) [\[Code\]](#)
- 6 **Yuhui Hong***, Mahsa Monshizadeh*, & Yuzhen Ye (2024). “Multitask Knowledge-primed Neural Network for Predicting Missing Metadata and Host Phenotype based on Human Microbiome.” *Bioinformatics Advances*, vbae203. [\[Paper\]](#) [\[Code\]](#)
- 7 **Yuhui Hong**, Sujun Li, Christopher J Welch, Shane Tichy, Yuzhen Ye, & Haixu Tang (2023). “3DMolMS: Prediction of Tandem Mass Spectra from Three Dimensional Molecular Conformations.” *Bioinformatics*, btad354. [\[Paper\]](#) [\[Code\]](#) [\[PyPI package\]](#) [\[Service on Konia\]](#)

PEER-REVIEWED ARTICLES (COLLABORATIVE AUTHOR)

- 8 Ludwig Lautenbacher, Kevin L. Yang, Tobias Kockmann, Christian Panse, Wassim Gabriel, Dulguun Bold, Elias Kahl, Matthew Chambers, Brendan X. MacLean, Kai Li, Fengchao Yu, Brian C. Searle, Wilburn, Damien, Mohammad Reza Zare Shahneh, **Yuhui Hong**, Haixu Tang, Mingxun Wang, Ralf Gabriels, Robbin Bouwmeester, Robbe Devreese, Tobias K. Schmidt, Alexey I. Nesvizhskii, & Mathias Wilhelm (2025). “Koina: Democratizing machine learning for proteomics research.” *Nature Communications*, 16(1), 9933. [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#)
- 9 Yaochen Li, **Yuhui Hong**, Yonghong Song, Chao Zhu, Ying Zhang, & Ruihao Wang (2022). “SiamPolar: Semi-supervised Realtime Video Object Segmentation with Polar Representation.” *Neurocomputing*, 467, 491-503. [\[Paper\]](#) [\[Code\]](#)
- 10 Yaochen Li, Chao Zhu, Yuehu Liu, **Yuhui Hong**, & Jianji Wang (2021). “Geometric and Semantic Analysis of Road Image Sequences for Traffic Scene Construction.” *Neurocomputing*, 465, 336-349. [\[Paper\]](#) [\[Code\]](#)

ONGOING ARTICLES

- 11 **Yuhui Hong**, & Haixu Tang (2025). “A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data.” (Under review). [\[Preprint\]](#) [\[Code\]](#)
- 12 **Yuhui Hong***, Mahsa Monshizadeh*, & Yuzhen Ye (2025). “Confounder Free Predictive Models for Microbiome-based Host Phenotype Prediction.” (Under review). [\[Preprint\]](#) [\[Code\]](#)

TEACHING EXPERIENCE

Course Designer DSCI-D590, AI on Ramp	Indiana University Bloomington From Fall 2025 onward
Instructor DSCI-D590, Topics in Data Science	Indiana University Bloomington Spring 2025
Instructor INFO-I529, Machine Learning Bioinformatics	Indiana University Bloomington Fall 2024
Assistant Instructor DSCI-D351, Big Data Analytics Instructor: Prof. Haixu Tang	Indiana University Bloomington Aug.-Sep. 2024

CONFERENCE PRESENTATIONS

- 1 **Invited talk** “Beyond database search: machine learning method for small molecule identification.”
College of Science, San Diego State University. Feb. 11, 2026. San Diego, CA.
- 2 **Oral Presentation** “A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data” [Slides]
73rd Conference on Mass Spectrometry and Allied Topics. Jun. 1 - 5, 2025. Baltimore, MD.
- 3 **Poster presentation.** “Predicting Compositional Fragments of Compounds from Their Tandem Mass Spectra Using Deep Neural Networks” [Poster]
72nd Conference on Mass Spectrometry and Allied Topics. Jun. 2 - 6, 2024. Anaheim, CA.
- 4 **Poster presentation.** “3DMolMS: Prediction of Tandem Mass Spectra from 3D Molecular Conformations”
Turkey Run Analytical Chemistry Conference 2023. Sep. 29 - 30, 2023. Marshall, IN.
- 5 **Oral Presentation** “A Machine Learning Model for Chemical Formula Prediction Using Tandem Mass Spectra of Compounds” [Slides]
71st Conference on Mass Spectrometry and Allied Topics. Jun. 4 - 8, 2023. Houston, TX.
- 6 **Poster Presentation** “Prediction of Molecular Tandem Mass Spectra Using 3-Dimensional Conformers” [Poster]
70th Conference on Mass Spectrometry and Allied Topics. Jun. 5 - 9, 2022. Minneapolis, MN.

SCHOLARSHIPS AND AWARDS

- **UW Data Science Fellow at the eScience Institute** 2025
University of Washington
Awarded \$1,000 fellowship for mass spectra data analysis
- **Luddy Outstanding Research Award** 2025
Indiana University Bloomington
Sole recipient university-wide for outstanding research contributions
- **Special Academic Scholarship** 2019
Xi'an Jiao Tong University
- **Meritorious Winner of MCM (Mathematical Contest In Modeling)** 2018
Consortium for Mathematics and Its Application (COMAP)
Top 10% internationally (ranked among 8,085 participating teams)

PROFESSIONAL SERVICES

- **Reviewer:** (conferences) KDD 2026, ACM BCB 2026 2025, EMBC 2026; (journals) Journal of Chromatography A, BMC Genomics, BMC Bioinformatics, IEEE Transactions on Computational Biology and Bioinformatics, PeerJ Computer Science, Pharmaceutical Research, Beilstein Journal of Organic Chemistry, Chemical Physics Letters.
- **Sub-reviewer:** (conferences) RECOMB 2026 2025 2023 2022, ISMB/ECCB 2025 2023, ACM BCB 2024; (journals) Analytical Chemistry, International Journal of Mass Spectrometry.
assisted in reviewing papers under the guidance of Professor William Noble and Professor Hairu Tang

PROFESSIONAL AFFILIATIONS

- American Society for Mass Spectrometry (ASMS), Member. 2022 - Present
- NSF Center for Bioanalytic Metrology (CBM), Student. 2022 - 2025

Last updated: April 2, 2026